

How a Naked Mole-Rat Queen's Scent Keeps the Peace

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Naked mole-rats live in colonies and work together. But they can also fight fiercely. Scientists recently found that a **queen** keeps her **colony** calm with a special scent.

In a **colony**, some mole-rats look for food and others care for the young. Only the **queen** has babies. When a **queen** dies, high-ranking mole-rats begin mating and fight hard until one becomes the new **queen**.

Gary Lewin is a **neuroscientist** in Berlin, Germany. His team found one chemical made only by **queens**. It is called **isopropyl myristate**. The team removed a **queen** from her **colony**. Then they put the chemical on the **colony** every day for three months. "In the entire three months, there was no fighting," Lewin said. When the researchers stopped adding the scent, the mole-rats began to fight. Within a few weeks, a new **queen** emerged.

The chemical also changes **hormones** in female mole-rats. The **queen** makes it only when she is pregnant or getting ready to be. The team shared its findings on July 15 in the **journal** *Nature*.

Word Treasure

colony -- A group of animals that live together.

queen -- The one female in a colony who has babies.

neuroscientist -- A scientist who studies the brain and nervous system.

isopropyl myristate -- The chemical made only by queen mole-rats.

hormones -- Chemicals in the body that send messages and control growth and behavior.

journal -- A magazine where scientists publish their research.

Break it down

Opening: a calming scent in naked mole-rat colonies

Life in a colony and what happens when a queen dies

The chemical and the three-month test

Hormones, when the queen makes the chemical and where the findings were shared

Why it matters

The study shows how animals can use chemicals to keep order in a group, which helps scientists understand how living things work together.

Test what you remember

1. Who is the only mole-rat that has babies in a colony?
 - (A) The scientists
 - (B) The food finders
 - (C) Any adult male
 - (D) The queen
2. Where does Gary Lewin work?
 - (A) Madrid, Spain
 - (B) Boston, United States
 - (C) Paris, France
 - (D) Berlin, Germany
3. How long did the team put the chemical on the colony?
 - (A) Five years
 - (B) Two weeks
 - (C) Three months
 - (D) One day
4. What happened when the scent was stopped?
 - (A) The queen came back
 - (B) The mole-rats went to sleep
 - (C) Nothing changed
 - (D) The mole-rats began to fight
5. What does the chemical change in female mole-rats?
 - (A) Their color
 - (B) Their teeth
 - (C) Their hormones
 - (D) Their size
6. When does the queen make the chemical?
 - (A) Only when she is sick
 - (B) Only when she is hungry
 - (C) When she is pregnant or getting ready to be
 - (D) All day, every day

Answer key (try the quiz first!): 1: D 2: D 3: C 4: D 5: C 6: C

MY THOUGHTS (at least 20 words):